

William Mattis

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Education

Bachelor of Science, Electrical Engineering

May 2026

University of Cincinnati – College of Engineering and Applied Science

Cincinnati, OH

- **Related Course Work:** Electromagnetics, Mechatronics, Digital Signal Processing, Embedded System

Experience

Bearcats Electric Racing (Electric FSAE Team)

Sept 2025 – Present

Electronics Team Lead

Cincinnati, OH

- Used excel to manage ordering of over 3,000 components for PCB assembly of team designed PCBs.
- Calculated and simulated thermal analysis of tractive system pre & discharge circuit using MATLAB.
- Spearheaded the creation of schematics, simulations, and layouts of PCBs for battery management operation including voltage monitoring, temperature monitoring, pre & discharge circuit for 600V powertrain system.
- Managed team of 7 seniors and underclassmen on PCBs for error indication of HV monitoring system as well as programmable reflow solder oven, and wiring harness for vehicle operation and data acquisition.
- Developed data acquisition system that was able to read and write CAN messages to assist in troubleshooting powertrain system as well as HaLow sub 1GHz Wi-Fi signal to transmit vehicle telemetry wirelessly.

X-Inator (Personal Start-Up)

June 2023 – July 2025

Co-Founder

Cincinnati, OH

- Collaborated with UC Venture Lab and other donors to raise over \$50,000 in non-diluted funding.
- Designed and manufactured glovebox humidity control loop, anode prototype molds, and high-pressure chamber (3000+ psi) for production of nanostructure material.
- Produced lab made coin cell batteries and imaged nanostructure material on scanning electron microscope.

Die-Craft Machine Shop

Jan – May 2023

Project Engineer

Cincinnati, OH

- Created CAD over 60 models and technical drawings for reference in 5 axis CNC manufacturing.
- Reversed engineered customer supplied parts for reproduction by machinists.

E-volve Systems

May – Aug 2022

Project Engineer

Blue Ash, OH

- Oversaw and validated installation of PLCs and control devices across a wide range of production facilities.
- Interfaced with system controls via PLC programming, plant management, and network management software.

Thermo Fisher Scientific

Jan – May & Aug – Dec 2021

THP Project Manager

Cincinnati, OH

- Oversaw installation, testing, and validation of over 20 temperature, humidity, and pressure monitoring systems in pharmaceutical production environments.
- Assisted in automating AutoCAD Electrical drawings for documentation on system installation.

Project Engineer

- Managed contractors on removal, construction, and installation of over 30,000 sqft of storage shelving.
- Conducted validation of installed pressure systems and electrical equipment in pharmaceutical production and development environments.

Skills

Software: Altium, MATLAB, Python, C++, LT Spice, NX, SolidWorks, Microsoft Office, Rockwell Automation

Hardware: PCB Design, Circuit Design & Analysis, Microcontrollers: (Arduino, ESP 32, Raspberry PI)

Manufacturing: PCB Manufacturing & Assembly, CNC Routing, CAD Modeling, 3D Printing

Campus Experience

Nanostructures Coursework

Jan – May 2024

- Utilized chemical processes to create graphene, nickel nanowires & gold nanoparticles.
- Measured the chemical makeup of various nanomaterials through Raman spectroscopy and UV spectrometry.
- Conducted visual analysis of carbon nanotubes, graphene, and nickel nanowires using scanning electron microscopes.